

Melanie Kung

Progressive positions leading to leadership roles in science, engineering, and technology.

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SUMMARY OF QUALIFICATIONS

Technical: materials testing, microfabrication, electrical circuits, analytical and computational methods

Languages: MATLAB, Python, JavaScript, Qiskit, VBA, HTML, CSS, LaTeX, C++, React

PROFESSIONAL EXPERIENCE

Research & Development Student, IPEX Technologies, Inc.

Sep 2025 – Apr. 2026

- Characterized the **physical, mechanical, chemical, thermal and rheological** properties in **PVC, CPVC, and ABS** compounds to support product development and material qualification.
- Led evaluation of integration of recycled material into PVC, identifying opportunities to support sustainability objectives and reduce material costs by **over \$70K USD**.
- Designed and executed **DOE-based experiments** to quantify the effect of additives on mechanical and rheological properties of PVC.
- Produced and characterized polymer specimens to evaluate material performance and properties using **impact testing, tensile testing, torque rheometry, DSC, TGA, and XRF**.
- Communicated experimental findings using technical reports and presentations to maximize customer satisfaction.

Materials Lab Assistant, Seaspan

Jan 2025 – Apr 2025

- Accurately prepared **100+** specimens and measured and reported data for mechanical testing using **Vickers hardness, Rockwell hardness, Charpy impact, and tensile testing** methods while maintaining compliance with laboratory procedures.
- Conducted **case study research** and mechanical testing to examine the effect of rewelding on steel.
- Created weld drawings using **AutoCAD** to illustrate welding, cutting, and testing parameters.
- Supported in the management and organization of the lab database including procedures, reports, and data using **Microsoft Office** tools.

Engineering Student, Canada Post

May 2024 – Aug 2024

- Applied **lean manufacturing** and **process management** principles to develop problem statements action plans for process engineering and streamline data collection.
- Developed **VBA** macros in Microsoft Excel to improve efficiency of data analytics.

EDUCATION

BASc University of Waterloo, Nanotechnology Engineering

2023 – 2028

- Dean's Honours List